# Patent Infringement Detection Analysis — US 10,613,742 B2 (Samsung)

**Patent:** "Method of Providing User Interaction with a Wearable Device and Wearable Device Thereof" **Assignee:** Samsung Electronics Co., Ltd. | **Inventors:** Samudrala Nagaraju; Surjeet Govinda Dash **Issued:** April 7, 2020 | **Priority:** April 22, 2014 **Analysis date / sources accessed:** May 22, 2026 **Disclaimer:** Technical analysis only. Not legal advice.

## TL;DR

- **No commercial smartwatch surveyed practices the core independent claims of US 10,613,742 B2.** The asserted independent claims (1, 16, 28, 29) require *automatic* determination of left vs. right hand using a **pressure sensor or vibration sensor** — and the industry universally relies on *manual user toggles* (Apple, Google/Fitbit, Garmin, Samsung, Huawei, Xiaomi, Amazfit, etc.) combined with accelerometer/gyroscope-based orientation, not pressure or vibration sensing, for this function.
- **The closest commercial hardware analog — Apple's "Force Touch" pressure sensor in Apple Watch Series 1 through Series 5 / SE first-gen** — was deprecated in watchOS 7 (announced June 23, 2020) and physically removed beginning with the Series 6 and SE (September 2020). Even when present, it was used solely for on-screen "firm press" UI gestures, never for handedness or wrist-orientation determination. Severity: **NONE** for current products; **LOW (historical)** for the legacy Force-Touch generations.
- **Practical recommendation:** Do not pursue litigation based on independent claims 1/16/28/29 against the surveyed wearables; the critical "pressure or vibration sensor → determine left/right hand" limitation is absent across the market. A narrower opportunity may exist around **Huawei Watch D / Watch D2**, which contain a genuine wrist-strap pressure sensor and inflatable airbag, *if* discovery reveals that pressure data is also used in any worn-position determination — but available public documentation indicates the pressure sensor is used only for oscillometric blood-pressure measurement.

## Key Findings

1. **The claim limitation "pressure sensor or vibration sensor" is dispositive.** Each independent claim requires receiving signals from a pressure sensor (claims 28, 29) or vibration sensor (claims 1, 16) and using those signals to determine left/right hand. Commercial wearables overwhelmingly use accelerometers, gyroscopes, optical PPG, ECG electrodes, and barometric altimeters for motion/orientation — not pressure-against-skin sensors and not vibration sensors as inputs.
2. **The handedness-detection feature is universally *manual* across the industry.** Apple Watch, Pixel Watch, Fitbit, Garmin, and Samsung's own Galaxy Watch all require the user to *select* their wrist (and, for Apple, the Digital Crown side) in setup or settings. Apple Support documentation ("Change the language and orientation on Apple Watch") instructs users to "Go to the Settings app on your Apple Watch. Tap General, then tap Orientation." Garmin's

process is equivalent: "In the Garmin Connect mobile app… select 'General,' and there should be an option to select 'Wrist Worn On.'" Fitbit's official manual likewise instructs: "Tap Wrist > Dominant or Non-dominant" — and a Fitbit moderator confirms: "it won't rotate the screen." Tizen's HumanActivityMonitor API exposes only GESTURE_WRIST_UP, snap, tilt, etc.; no left/right-wrist auto-detect. Apple + 3

3. **Apple's Force Touch is the only "pressure sensor in the watch body" actually shipped in a major smartwatch line**, but its function was screen-pressure UI (firm press) — implemented around the display with an Analog Devices AD7149 capacitance-sensor controller (confirmed by iFixit's Apple Watch Series 2 Teardown, Sept. 15, 2016, and corroborated by TechInsights' Series 2 teardown: "the same Analog Devices AD7149 capacitance sensor controller we have observed in previous generations of Apple Force Touch-enabled devices"). It was never used as a worn-position sensor. It was retired in 2020 (9to5Mac, June 23, 2020: "watchOS 7 drops Force Touch support"). TechInsights 9to5Mac
4. **Huawei Watch D / D2** contains a genuine pressure sensor and inflatable airbag in the strap — a configuration superficially close to dependent claims 3–5 ("a pressure sensor configured at a strap"). However, the sensor is used for oscillometric blood-pressure measurement, not wrist-side or worn-position determination — so independent-claim limitations are not met. Huawei's official D2 page describes a "TruSense system… bolstered by a high-precision pressure sensor, mini pump, and inflatable airbag, to provide precise BP measurements." HUAWEI
5. **Vibration sensors (claims 1, 16) are essentially absent from commercial smartwatches.** All major watches contain a vibration *motor* (Apple's Taptic Engine, generic linear resonant actuators) for haptic *output*, but no surveyed device uses it as an *input* sensor (e.g., measuring back-EMF) to determine the hand of wear. No relevant Tizen, Wear OS, or watchOS public API exposes this.
6. **Dependent claims requiring sensors in BOTH body AND strap (claims 3–5, 6–8, 19–24)** narrow the field further. No surveyed product has documented pressure or vibration sensors in both the body and the strap arranged for the claimed comparison logic. Apple's older Smart Connector pins and current strap latches are electrical/mechanical only.
7. **No known litigation on US 10,613,742 B2 or its parent US 10,275,143.** Public dockets and law-firm press releases show no infringement suit on this specific patent. Samsung has been a defendant in several recent unrelated patent matters — most notably **Pictiva Displays International Ltd. v. Samsung** (E.D. Tex. 2:23-cv-495; $191.4M jury verdict Nov. 3, 2025) involving OLED patents asserted against "Samsung's Galaxy smartphones, televisions, computers, [and] wearables" (Reuters / McKool Smith), which is the only recent verdict actually touching Samsung wearables. **Maxell Ltd. v. Samsung** ($111.7M jury verdict May 28, 2025; subsequently overturned by the trial judge on Sept. 19, 2025 per Law360) concerned device-unlock/imaging patents in smartphones and home devices, not wearable handedness. **Anonymous Media Research Holdings v. Samsung** (E.D. Tex. 2:23-cv-00439; $78.5M verdict Oct. 6, 2025) concerned automatic content recognition in smart TVs and is not wearables-adjacent. None of these touch the '742 patent family. U.S. News & World Report + 2

## Details (Product-by-Product Analysis)

> **Methodology note:** Each entry lists (a) the relevant claim limitations, (b) the product's actual function, (c) the sensors used (pressure, vibration, accelerometer, gyroscope, PPG, etc.), (d) whether wrist-detection / orientation is automatic or user-set, and (e) the resulting severity. "Literal infringement" means each claim limitation is met as written; "doctrine of equivalents" (DoE) means the accused element performs substantially the same function in substantially the same way with substantially the same result.

### 1. Apple — Apple Watch (all generations)

- **Functionality:** Manual "Watch Orientation" setting in iOS Watch app or watchOS Settings → General → Orientation. User selects Left/Right wrist and Left/Right Digital Crown side. Display flips 180° based on this manual selection. Apple Support (apple.com/en-sg/guide/watch/apd0bf18f46b/watchos): "Go to the Settings app on your Apple Watch. Tap General, then tap Orientation."
- **Sensors used for orientation:** Accelerometer + gyroscope (not pressure/vibration). Apple's "Force Touch" capacitive pressure sensor (Series 1–5 and SE first-gen) was used for on-screen firm-press UI only and never read to determine handedness; disabled in watchOS 7 (Sept 2020) and physically removed beginning Series 6 / SE (Sept 2020). iFixit's Apple Watch Series 6 Teardown (Sept 21, 2020): "The procedure is also slightly simplified due to the absence of a Force Touch gasket—a feature now defunct as of watchOS 7." MacRumors: "Apple has removed all Force Touch interactions from the UI, making the Force Touch sensor gasket in Apple Watch Series 5 and earlier models effectively superfluous." 9to5Mac MacRumors
- **Claim mapping:**
  - Claims 1, 16 (vibration sensor → determine hand): **NOT MET.** No vibration sensor used as input.
  - Claims 28, 29 (pressure sensor → determine hand): **NOT MET.** Force Touch existed only in older models, used for screen-press UI, not handedness; Series 6+ have no Force Touch hardware at all.
  - Claims 2, 17 (gyroscope for orientation): would map literally if a primary claim were met; no primary claim is met.
  - Claim 15 (swipe-gesture call accept/reject differing by hand): Apple Watch's call accept/reject gestures are the same regardless of wrist; the display flips but the gesture mapping is not asymmetric by hand.
- **Severity: NONE** for current Apple Watch (Series 7, 8, 9, 10, SE 2, Ultra, Ultra 2). **LOW (historical)** for Series 1–5 only under a strained DoE theory that Force Touch's capacitive sensing equates to a "pressure sensor" — but the claim still fails because the sensor was never used to determine wrist side.
- **Confidence: Very High.**

**2. Google / Fitbit — Pixel Watch 1–4; Fitbit Sense, Versa 4, Charge 6**

- **Functionality:** Pixel Watch and Fitbit devices require manual "Wrist" (Dominant / Non-dominant) selection. The setting tunes step-counting sensitivity; Google/Fitbit Help confirms: "To start, the Wrist setting is set to non-dominant. Change your wrist setting in the Fitbit app." A Fitbit moderator confirms: "The setting above will tell the system where are you wearing your tracker so it knows if it is on the dominant or non dominant hand but it won't rotate the screen." Google Support + 2
- **Sensors:** Accelerometer, gyroscope, barometric altimeter (used for floor counting via atmospheric pressure changes, not for handedness), PPG, ECG, skin temperature, cEDA. No vibration sensor as input.
- **Claim mapping:** Claims 1, 16, 28, 29: **NOT MET.** No vibration sensor; no pressure sensor used for hand detection; even the manual setting only modulates step-count sensitivity.
- **Severity: NONE.**
- **Confidence: Very High.**

**3. Garmin — Forerunner, Fenix, Venu, Vivoactive, Epix, Instinct**

- **Functionality:** Manual "Wrist Worn On" setting in Garmin Connect (Garmin Devices → [device] → General). Garmin watches have asymmetrical physical buttons, so the display is not rotated by software. A Garmin Forum verified post (vivoactive 4 series): "In Connect App tap on the top watch image, go to general and change the 'Wrist worn on' option." Garmin Forums
- **Sensors:** Accelerometer, gyroscope, barometric altimeter, PPG, GPS. No vibration-sensor input. No body-pressure sensor on skin.
- **Claim mapping: NOT MET** on any independent claim. The setting is purely manual; affects algorithm tuning and a few backlight-gesture mappings only.
- **Severity: NONE.**
- **Confidence: Very High.**

**4. Samsung — Galaxy Watch (Patent Owner, included only for baseline)**

- Samsung is the assignee; its own products cannot infringe. Galaxy Watch 4/5/6/7/Ultra expose a manual orientation setting (SamMobile "Turn your Samsung Galaxy Watch 180 degrees!"), suggesting Samsung itself has not commercialized the automated pressure/vibration-based detection of the patent in its retail products. This bears on commercial significance / damages base if the patent were ever enforced against third parties.
- **Severity:** N/A (assignee).

**5. Huawei — Watch GT, Watch 4, Watch D, Watch D2, Watch Ultimate; Band series**

- **Functionality:** Huawei Watch D2 includes a genuine **high-precision pressure sensor +**

**mini pump + inflatable airbag** in the strap for oscillometric blood-pressure measurement. Huawei's official D2 page: "brand-new TruSense system is bolstered by a high-precision pressure sensor, mini pump, and inflatable airbag." TechRadar review of the Watch D: "the Watch D incorporates an actual blood pressure cuff inside the watch band." HUAWEI TechRadar

- **Critical point:** While this hardware aligns superficially with dependent claim 4 ("a pressure sensor configured at a strap"), the sensor's use is BP measurement only — public documentation does not indicate it is used to determine wrist side or worn position. The independent claims (28, 29) require the pressure sensor's output to be used to determine left vs. right hand, which Huawei does not document doing.
- **Claim mapping:**
  - Claim 28/29: **NOT MET** literally because the sensor is not used for handedness determination.
  - Dependent claims requiring pressure sensor in strap (4, 20): partially met *physically* but inoperative because no independent claim is met.
  - Claims 1/16 (vibration sensor): **NOT MET.**
- **Severity: LOW (contingent).** If discovery reveals Huawei uses the pressure signal to validate worn-position (e.g., verifying the watch is on a wrist before BP measurement) and that this implicates handedness, the analysis could move to **MEDIUM** for Watch D / D2 specifically. Public documentation does not support that escalation today.
- **Confidence: Medium.**

**6. Xiaomi (Mi Watch, Mi Band, Redmi Watch), Amazfit/Zepp/Huami (GTR, GTS, T-Rex, Stratos), OnePlus Watch, Oppo Watch, Fossil Gen 5/6, TicWatch/Mobvoi Pro**

- **Functionality:** These Wear OS, Zepp OS, RTOS, or proprietary-OS devices universally use accelerometer/gyroscope motion. None publicly document automatic left/right detection. Most do not even offer a wrist-orientation toggle (display is fixed). iFixit's Xiaomi Mi Watch 2021 Teardown identifies only standard MCU, PPG, and motion components — no pressure or vibration input sensor.
- **Sensors:** Accelerometer, gyroscope, PPG, sometimes SpO2, some barometric altimeters.
- **Claim mapping: NOT MET** across the board.
- **Severity: NONE.**
- **Confidence: High.**

**7. Polar (Vantage, Grit X, Ignite), Suunto (7, 9), Withings (ScanWatch, Steel HR), Casio (WSD series, G-Shock smart)**

- **Functionality:** Sport/health watches; manual wrist settings where applicable; no documented automatic wrist-side detection. Accelerometer/gyroscope based.
- **Severity: NONE.**
- **Confidence: High.**

**8. Related (non-infringing) automatic-detection patent**

- USPTO-published US 11,630,120 ("System and method for detecting wrist-device wearing location") describes **automatic detection of left vs. right limb using an accelerometer with threshold comparison**, reciting "an accelerometer... compare each of the plurality of accelerometer samples to a first threshold... determine that the apparatus is either associated with a left limb or a right limb based on the comparison." This patent's named assignee could not be independently confirmed from public sources and should be verified before any prior-art use — but its existence is significant: it shows a *design-around* of US 10,613,742 using an accelerometer (a sensor type that '742's specification explicitly distinguishes from pressure/vibration sensors). With respect to '742: **NONE** (different sensor class). uspto

**9. Mudra Band / Mudra Link (Wearable Devices Ltd.)**

- **Functionality:** Surface Nerve Conductance (SNC) / EMG wristband with IMU (per Mudra-band.com and Hackster.io coverage of CES 2025: "captures and analyzes neural patterns" from "three Surface Nerve Conductance (SNC) sensors" plus a "6DoF IMU"). Used to detect finger gestures via bioelectric signals. Mudra Wearable Inc + 2
- **Claim mapping:** SNC sensors detect bioelectric activity, not pressure or vibration. The device is a gesture controller, not a self-rotating display device, so the orientation/handedness-display claims do not apply.
- **Severity: NONE.**
- **Confidence: High.**

**10. Whoop, Oura Ring**

- Whoop is a strap-form fitness tracker with no display; UI re-layout and gesture-based call-accept claims are non-applicable. Oura is a ring, not a wrist device, and is therefore outside the literal scope of "a wrist on which the wearable device is being worn."
- **Severity: NONE.**

**11. Honor, Realme, Noise, boAt, Fire-Boltt, Imilab (Chinese / Indian budget brands)**

- Predominantly accelerometer/gyroscope/PPG-only devices, often with fixed-orientation displays. No documented automatic wrist-side detection.
- **Severity: NONE. Confidence: High.**

## Recommendations

**Tier 1 — Do not pursue (NONE/LOW severity):** Apple Watch (current), Pixel Watch / Fitbit, Garmin, Xiaomi, Amazfit, OnePlus, Oppo, Fossil, TicWatch, Polar, Suunto, Withings, Casio, Honor, Realme, Noise, boAt, Fire-Boltt, Imilab, Mudra, Whoop, Oura. The "pressure or vibration sensor → automatic left/right hand determination" limitation is absent. Even under DoE, manual user toggle is not "substantially the same way" as sensor-based automatic determination, defeating the function-way-result test.

**Tier 2 — Targeted investigation (potential LOW–MEDIUM):**

1. **Huawei Watch D / Watch D2.** Conduct an FRCP-11-compliant pre-suit investigation to determine (a) whether the wrist-strap pressure sensor's signal is used at any firmware stage to determine handedness or worn position (not just BP measurement), and (b) whether the airbag-inflation sequence implicitly validates worn position in ways implicating left/right detection. **Escalation threshold:** documentary evidence (firmware analysis, source code, technical white paper, or sworn declaration) that pressure-sensor output is fed into a worn-position or handedness classifier. If yes → **MEDIUM**; if no → **NONE**.
2. **Historical Apple Watch Series 1–5 / SE first-gen (pre-Sept 2020).** Statute of limitations is six years pre-suit under 35 U.S.C. § 286; the product line is discontinued and the Force Touch sensor was undisputedly used only for screen-press UI. Recommend not pursuing absent new evidence (e.g., Apple service-mode diagnostics using the sensor for worn-position).

**Tier 3 — Patent strategy alternatives:**

- A DoE argument equating accelerometer-based wrist detection with the claimed vibration sensing is likely a losing argument: the patent specification explicitly contrasts pressure/vibration with accelerometer/gyroscope, and adding a gyroscope in dependent claim 2 indicates the patentee viewed them as distinct categories — implicating disclosure-dedication and claim-vitiation doctrines.
- Evaluate Samsung's continuation family for any pending continuations that could be prosecuted to cover accelerometer-based handedness detection (without running afoul of written-description requirements).
- Consider licensing rather than litigation; many wearable makers may take a nominal license to avoid litigation costs even on a weak theory, and this can establish a royalty base for future negotiations.

**Benchmarks that would change the recommendation:**

- Discovery of any firmware/SDK that uses vibration-motor back-EMF as a sensor input → reopen analysis (potentially Apple Taptic Engine or Samsung vibrator, but no public evidence).
- A new product launch using strap-embedded pressure sensors for wrist verification → trigger re-analysis.
- A USPTO continuation issuing with broader claims (e.g., covering accelerometer-based determination) → expand enforcement universe.

## Caveats

1. **Patent validity risks.** US 10,613,742 has a priority date of April 22, 2014 and issued April 7,

2020. Prior art for accelerometer/gyroscope-based handedness detection (e.g., the academic literature on activity recognition by wrist side from 2013–2016, MDPI Sensors 16(6):800 and similar) is dense. The patent was likely allowed because it is narrowed to *pressure or vibration sensors* — but that narrowing also limits enforceability. Subject to potential IPR or 35 U.S.C. § 112 challenges if Samsung tries to broaden via continuation.

2. **Claim construction ambiguity.** "Vibration sensor" is not narrowly defined in the spec. A defendant could argue accelerometers detect vibration (since high-frequency accelerometer data captures micro-vibrations). However, the spec consistently distinguishes the two, and dependent claim 2 adds a gyroscope as a separate sensor — supporting the narrow reading.
3. **Doctrine-of-equivalents limits.** Function-way-result generally fails because manual user selection of wrist (the universal industry approach) is not "substantially the same way" as automatic sensor-based determination. Prosecution-history estoppel may further limit DoE.
4. **Confidentiality and discovery limits.** This analysis relies on publicly available teardowns, datasheets, marketing, and developer documentation. Internal firmware (Apple S-series SoC, Huawei TruSense, various Chinese OEM firmwares) may use sensors in ways not publicly disclosed. A complete investigation would require subpoenaed source code.
5. **Geographic and product scope.** US 10,613,742 is a U.S. patent only. Samsung has parallel filings (WO 2015/163653 A1 and national counterparts); foreign enforcement requires jurisdiction-specific claim-scope analysis (e.g., Germany's "central means" doctrine differs from U.S. DoE).
6. **No litigation history.** Absence of prior assertion suggests Samsung has either (a) not viewed the patent as commercially significant for offensive enforcement, or (b) reserved it for defensive/cross-licensing. This is consistent with most large electronics OEMs' wearable patent portfolios.
7. **Information accessed May 22, 2026; product features may change post-analysis** — particularly Huawei Watch D2 firmware updates, Apple watchOS feature additions, and any new Samsung continuations or assertions.
8. **This analysis is technical only and does not constitute legal advice.** A registered patent attorney must perform formal claim construction, opinion-of-counsel, and litigation-feasibility analysis before any enforcement action.

## Summary Table

| Company / Product Family | Vibration Sensor (input)? | Pressure Sensor (body)? | Pressure Sensor (strap)? | Auto Left/Right Detect? | Display Auto-Rotate? | Severity | Confidence |
|---|---|---|---|---|---|---|---|
| Apple Watch (Series 7+, | No | No | No | No (manual) | Manual setting | NONE | Very High |

| Company / Product Family | Vibration Sensor (input)? | Pressure Sensor (body)? | Pressure Sensor (strap)? | Auto Left/Right Detect? | Display Auto-Rotate? | Severity | Confidence |
|---|---|---|---|---|---|---|---|
| Ultra, Ultra 2, SE 2) | | | | | | | |
| Apple Watch (Series 1–5 / SE first-gen) | No | Force Touch (AD7149, screen UI only) | No | No (manual) | Manual setting | LOW (historical) | Very High |
| Pixel Watch 1–4 | No | No (altimeter is barometric only) | No | No (manual) | No rotate | NONE | Very High |
| Fitbit Sense / Versa 4 / Charge 6 | No | No (altimeter only) | No | No (manual) | No rotate | NONE | Very High |
| Garmin (Forerunner / Fenix / Venu / Vivoactive / Epix / Instinct) | No | No | No | No (manual) | No rotate | NONE | Very High |
| Samsung Galaxy Watch (assignee) | N/A | N/A | N/A | Manual | Manual | N/A (owner) | Very High |
| Huawei Watch GT / 4 / Ultimate | No | No | No | No (manual) | Manual setting | NONE | High |
| Huawei Watch D / Watch D2 | No | No (BP optical separately) | YES (BP airbag) | No (manual) | Manual setting | LOW (contingent) | Medium |
| Xiaomi Mi Watch / Mi Band / | No | No | No | No | Varies | NONE | High |

| Company / Product Family | Vibration Sensor (input)? | Pressure Sensor (body)? | Pressure Sensor (strap)? | Auto Left/Right Detect? | Display Auto-Rotate? | Severity | Confidence |
|---|---|---|---|---|---|---|---|
| Redmi Watch | | | | | | | |
| Amazfit (Zepp / Huami) | No | No | No | No | No | NONE | High |
| OnePlus Watch 1/2 | No | No | No | No | No | NONE | High |
| Oppo Watch | No | No | No | No | No | NONE | High |
| Fossil Gen 5/6 | No | No | No | No | No | NONE | High |
| TicWatch Pro (Mobvoi) | No | No | No | No | No | NONE | High |
| Polar Vantage / Grit X / Ignite | No | No | No | No (manual) | No | NONE | High |
| Suunto 7 / 9 | No | No | No | No (manual) | No | NONE | High |
| Withings ScanWatch / Steel HR | No | No | No | No | No | NONE | High |
| Casio WSD / G-Shock smart | No | No | No | No | No | NONE | High |
| Honor / Realme / Noise / boAt / Fire-Boltt / Imilab | No | No | No | No | No | NONE | High |
| Mudra Band / Link | No (EMG/SNC) | No | No | No | N/A (no display) | NONE | High |

| Company / Product Family | Vibration Sensor (input)? | Pressure Sensor (body)? | Pressure Sensor (strap)? | Auto Left/Right Detect? | Display Auto-Rotate? | Severity | Confidence |
|---|---|---|---|---|---|---|---|
| Whoop | No | No | No | N/A | N/A (no display) | NONE | High |
| Oura Ring | No | No | No | N/A (ring) | N/A | NONE | High |

**Final assessment:** US 10,613,742 B2's specific claim limitations — pressure or vibration sensors used to automatically determine wrist side — are not practiced by any of the surveyed commercial wearables as of May 22, 2026. The patent reads on a technical approach that the industry has not adopted, instead converging on (a) manual user configuration plus (b) accelerometer/gyroscope-based orientation. Without significant claim broadening (which prosecution history likely forecloses) or discovery of undisclosed firmware behavior, the patent's enforceable scope against commercial smartwatch makers is narrow. The single product family meriting further investigation is the Huawei Watch D / D2 because of its strap-mounted pressure sensor — but on present evidence, that pressure sensor is used only for blood-pressure measurement, not for the handedness or worn-position determination required by the independent claims.